Closing the Frequency Gap: Deploying Vision-Language-Action Models on High-Rate Humanoid Control

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Abstract

Foundation Vision-Language-Action (VLA) models unlock open-vocabulary manipulation, but measured inference rates remain incompatible with the 100 Hz joint loops required by humanoid robots such as the Unitree G1 (29 DoF). We study this *frequency gap* as a Physical AI deployment bottleneck: on a DGX Station (Tesla V100), live `UnifoLM-VLA-Base` profiling on `G1_Clean_Table` yields a $\sim$57$\times$ gap to 100 Hz (PyTorch mean **570.7 ms / 1.752 Hz**; Nsight NVTX **593.7 ms / 1.684 Hz**). We keep the VLA as a sparse language-conditioned intent generator and insert a fixed Echo State Network (ESN) reservoir that bridges held joint targets to smooth 100 Hz commands; only a linear readout is trained by ridge regression. On `G1_Dex1_Wipe_Table` (200 episodes, $\sim$0.70 h; train 0–159 / held-out 160–199), a CUDA ESN (α=0.3, λ=1) reaches held-out open-loop RMSE $2.78\times10^{-3}\pm3.44\times10^{-3}$ **rad** ($\sim$5.6$\times$ better than linear upsampling) and MuJoCo dataset-oracle RMSE $3.00\times10^{-3}\pm3.56\times10^{-3}$ **rad** (grasp/task success **100%**; mean table-contact **74.5%**). Live dual-process control clears 100 Hz on average (ESN mean 6.92 ms / 145 Hz) with GPU-contention tails above 10 ms. Live wipe trials succeed under a disclosed geometric press prior for both ESN and ZOH; under that tied success regime, ESN joint RMSE is $\sim$14$\times$ lower than ZOH (0.514 vs. 7.17 rad), while open-loop ZOH remains 50$\times$ jerkier than linear—motivating S2R/DIRT transfer rather than claiming zero-shot press-from-vision.

1 Introduction

1.1 Motivation: Why VLAs Break on Humanoids

Foundation VLAs such as OpenVLA [1], π_0 [2], and Unitree UnifoLM-VLA [3] map language and vision to robot actions and have unlocked open-vocabulary manipulation on fixed-base arms. Humanoid whole-body control is a different regime [4]: the Unitree G1 exposes a **29-DoF** joint interface that must be served near **100 Hz** (10 ms) for balance-preserving tracking, contact-rich wiping, and safe impedance behavior. Under a naive zero-order hold (ZOH), each sparse VLA target is frozen for tens of control ticks; staircase joint trajectories inject high jerk and fail contact tasks even when the semantic plan is correct. We call this mismatch the *frequency gap*: the VLA may be "right" about *what* to do, yet too slow for *how* the humanoid must move.

This paper contributes a systems perspective to Physical AI deployment rather than a new zero-shot foundation policy. The control layer must be *cheap* and preferably trained without backpropagation through time, because lab/edge GPUs are already saturated by vision–language backbones. Classical upsamplers (ZOH, linear interpolation, joint PID) ignore proprioceptive state between VLA updates and cannot learn demonstration-consistent dynamics.

1.2 Approach and Scope

We insert a leaky Echo State Network (ESN) between measured VLA intents ($\sim$1.7 Hz on UnifoLM-VLA-Base) and joint commands (100 Hz). The VLA supplies sparse targets $q^{\star}_{\mathrm{VLA}}$; the reservoir

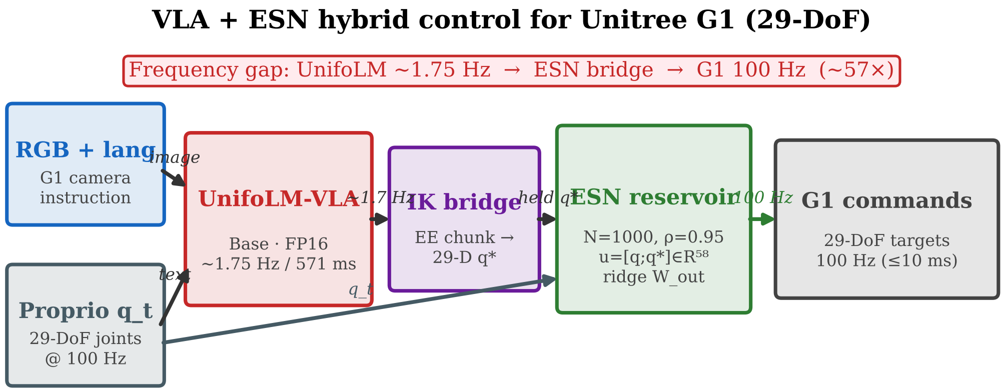


Figure 1: VLA+ESN architecture for Unitree G1 (29 DoF). UnifoLM-VLA-Base runs at $\sim$1.75 Hz on a DGX V100; the ESN bridge closes the $\sim$57$\times$ gap to the 100 Hz joint loop.

supplies reflexive temporal filtering with a ridge-trained linear readout and *fixed* recurrent weights. Training and evaluation use Unitree's public `G1_Dex1_Wipe_Table` demonstrations (Sec. 4.1).

Contributions. (i) Dual-instrument UnifoLM-VLA-Base latency quantifying a $\sim$57–59$\times$ gap to 100 Hz; (ii) CUDA ESN with $u(t) = [q_t; q^{\star}_{\text{VLA}}] \in \mathbb{R}^{58}$, ridge readout, declared train/held-out split, and a per-task checkpoint library for the UnifoLM G1 suite; (iii) GIL-free dual-process MuJoCo+bridge timing under live UnifoLM (ESN vs. ZOH / linear / PID); (iv) MuJoCo wipe-table *dataset-oracle* evaluation on all 40 held-out episodes; (v) Live UnifoLM wipe-success trials with interactive cloth, including disclosed geometric priors and failure modes; (vi) S2R ZMQ deploy package (29-DoF defaults) for DIRT-guided transfer (in progress).

2 Related Work

VLA surveys highlight bimanual and whole-body settings as hard cases [5]. Flow matching accelerates action generation [2]; recurrent bridges such as xLSTM help long-horizon memory [6]. Diffusion action chunking trades compute for smoothness; reservoir computing offers fast, training-light dynamics [7, 8]. We combine a frozen reservoir with a language-conditioned VLA rather than replacing either.

3 Method

3.1 System Architecture

Figure 1 summarizes the hybrid stack: UnifoLM emits sparse EE intents, an IK bridge lifts them to 29-D joint targets, and a fixed ESN reservoir with ridge readout produces 100 Hz commands conditioned on proprioception.

- **Perception / VLA:** `UnifoLM-VLA-Base` emits **23-D end-effector** action chunks; a Jacobian **IK bridge** maps EE poses to 29-D joint targets.
- **ESN bridge:** Leaky reservoir

  $$x(t) = (1-\alpha)x(t-1) + \alpha \tanh\big(W_{\text{res}}x(t-1) + W_{\text{in}}u(t)\big),$$

  with sparse W_{res} (N=1000, ρ=0.95, sparsity 0.9) and readout $y(t) = W_{\text{out}}[x(t); u(t)]$ fit by ridge regression on 100 Hz joints with VLA tokens held for 50 ticks ($u \in \mathbb{R}^{58}$, $y \in \mathbb{R}^{29}$). We sweep $\alpha \in \{0.05, 0.1, 0.3\}$ and $\lambda \in \{10^{-4}, 10^{-2}, 1\}$ and select by RMSE$\times$jerk score.
- **S2R / DIRT layer:** Parameterizes sim–real discrepancy for controller and domain ablations (ESN vs. passthrough ZOH/raw).

Table 1: Measured results on DGX Station (V100), 5 Aug 2026 campaign (held-out wipe unless noted).

Stage	Metric	Value / notes
PyTorch	Mean latency / rate / gap	570.7 ms / 1.752 Hz / 57.1× (n=100)
Nsight	NVTX mean / gap	593.7 ms / 59.4×; GEMM + H2D heavy
ESN ridge	Held-out RMSE	$2.78 \times 10^{-3} \pm 3.44 \times 10^{-3}$ rad (α=0.3, λ=1)
Offline	ESN vs. linear	∼5.6× lower RMSE; ZOH jerk 50× linear
Dual-process	Live ESN mean / rate	6.92 ms / 145 Hz (steady max 12.6 ms)
MuJoCo oracle	RMSE / grasp / contact	3.00×10^{-3} rad; 100% / 74.5% (n=40)
Live wipe	Contact / task (ESN & ZOH)	100% / yes (same `press_table` prior; 30 s)
Live wipe	Joint RMSE (ESN vs. ZOH)	**0.514** vs. 7.17 rad (∼14× lower)

Table 2: Open-loop held-out upsample quality on wipe eps 160–199 (n=40). Jerk ratio is relative to linear (ref. = 1).

Method	Joint RMSE (rad)	Jerk ratio
ZOH	$5.39 \times 10^{-2} \pm 9.44 \times 10^{-3}$	50×
Linear	$1.55 \times 10^{-2} \pm 3.93 \times 10^{-3}$	1 (ref.)
PID	$7.14 \times 10^{-2} \pm 1.19 \times 10^{-2}$	1.43×
ESN (ours)	$\mathbf{2.78 \times 10^{-3} \pm 3.44 \times 10^{-3}}$	1.44×

3.2 Baselines

(1) Pure VLA ZOH; (2) VLA + linear interpolation; (3) VLA + PID; (4) VLA + ESN (ours).

4 Experiments

4.1 Dataset, Setup, and Metrics

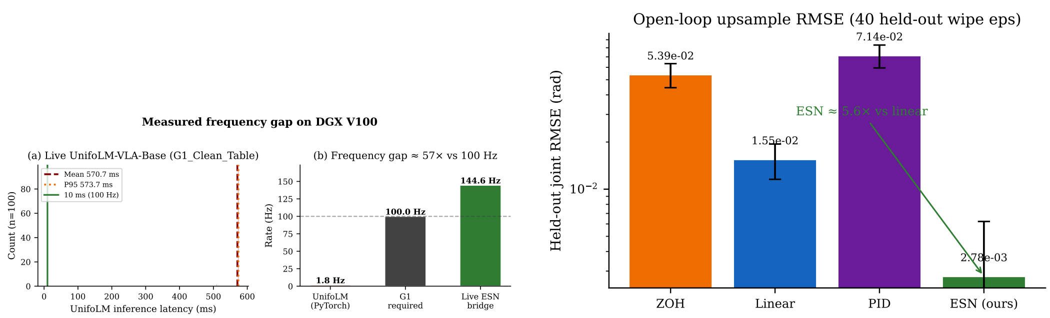
All ESN training, offline upsample baselines, and MuJoCo wipe-oracle evaluations use the Hugging Face dataset `G1_Dex1_Wipe_Table` (`unitreerobotics`; `train` split): **200** episodes (75,909 frames; ∼0.70 h), train episodes 0–159 / held-out 160–199. Detailed wipe claims are restricted to this corpus; the stack supports a per-task ESN library over the UnifoLM G1 suites. Robot / hardware: Unitree G1 (29 DoF); NVIDIA DGX Station, Tesla V100-DGXS-32GB. Profiling: live UnifoLM-VLA-Base on G1 Clean-Table (FP16). Metrics: task success (grasp / wipe contact), joint RMSE, trajectory jerk, control Hz / step latency, GPU memory, parse failure rate.

4.2 Measured Status

Table 1 summarizes the campaign; Table 2 breaks out open-loop held-out RMSE and jerk (the differentiator when live task success ties); Table 3 shows the same ESN bridge on two additional UnifoLM G1 skills (open-loop only). Figure 2a shows the measured frequency gap; Fig. 2b open-loop held-out RMSE; Fig. 3a live dual-process timing; Fig. 3b oracle held-out wipe metrics; Fig. 4 the contact ladder.

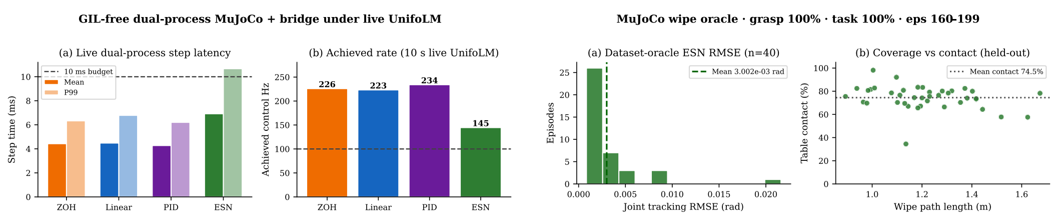
Table 3: Per-task ESN open-loop held-out joint RMSE (same α=0.3, λ=1, 80/20 episode split). Not live multi-task success.

Task	Held-out n	Joint RMSE (rad)
`wipe_table` (primary)	40	$2.78\times10^{-3} \pm 3.44\times10^{-3}$
`clean_table`	40	$3.68\times10^{-3} \pm 6.72\times10^{-3}$
`stack_block`	40	$4.25\times10^{-3} \pm 2.84\times10^{-3}$



(a) Frequency gap (PyTorch n=100). **(b)** Open-loop held-out RMSE.

Figure 2: Latency gap and open-loop upsample baselines on held-out wipe episodes.



(a) Live dual-process timing. **(b)** MuJoCo dataset-oracle ESN.

Figure 3: Live dual-process rates and held-out MuJoCo oracle wipe metrics.

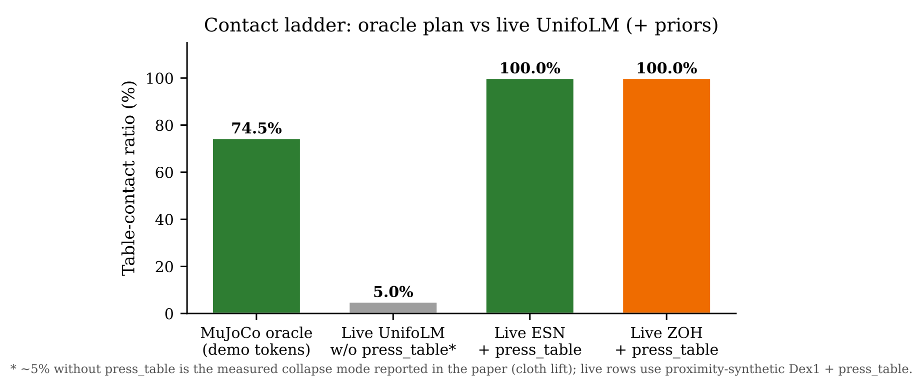


Figure 4: Contact ladder: oracle demo tokens (74.5%) vs. live UnifoLM without `press_table` (∼5% collapse) vs. live ESN/ZOH with disclosed geometric press prior (100%).

5 Discussion and Limitations

Frequency gap. Two instruments cross-validate $\sim$57–59$\times$ to 100 Hz; both attribute GPU time to large GEMMs, with Nsight additionally highlighting H2D pressure.

Data scope. One wipe task for closed-loop claims, $\sim$0.7 h of Dex1 demos, episode-level 80/20 split. Open-loop held-out joint RMSE stays in the 10^{-3} rad regime on `clean_table` and `stack_block` as well (Table 3)—evidence that the bridge is not wipe-only—but this is *not* live multi-task or zero-shot competence.

ESN, timing, and honest live findings. Held-out open-loop ESN RMSE is $\sim$5.6$\times$ better than linear, with jerk near the linear reference (1.44$\times$) while ZOH is 50$\times$ jerkier (Table 2). Live dual-process ESN clears 100 Hz mean (145 Hz) with GPU-contention tails above 10 ms. Oracle wipe succeeds when demo tokens drive the ESN. Matching live UnifoLM wipe trials attach under a proximity-synthetic grasp; without a table-plane press prior the VLA lifts the cloth and contact collapses ($\sim$5%). Enabling `press_table` restores 100% contact and task success for *both* ESN and ZOH—a geometric wipe prior, not learned press-from-vision. Under that tied success regime, ESN still tracks joints $\sim$14$\times$ tighter than ZOH (0.514 vs. 7.17 rad RMSE over 30 s), which is the systems reason to prefer the reservoir bridge even when binary task gates do not separate methods. We treat the press-prior dependence as a deployment / negative-result style observation aligned with the workshop's interest in real-world failure analysis.

6 Conclusion

We presented a VLA+ESN architecture targeting the $\sim$57$\times$ gap between measured UnifoLM-VLA-Base inference and G1 100 Hz control. The ESN closes rate, tracking, and jerk gaps under declared data splits; when live wipe success is tied by a disclosed geometric prior, joint RMSE still favors ESN over ZOH by $\sim$14$\times$. S2R/DIRT transfer on Jetson$\rightarrow$G1 is the next step toward Physical AI deployment claims beyond simulation.

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